STAT 566: Causal Modeling

Note 01: Potential Outcomes, Randomized Experiments, and Partial Identification

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Reading guide

English transcription and edited study notes. Yellow boxes preserve manuscript TODOs; red boxes explain corrections; blue boxes identify lecture-based additions. Gray boxes clarify questions. Source-page markers refer to the original handwritten PDF, not this edition.

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1 Potential outcomes and causal estimands

Manuscript: stat566-note01, pp. 1–2 (April 2).

Let $A \in \{0, 1\}$ denote treatment and $Y(a)$ the potential outcome under treatment a. The observed outcome is $Y = Y(A)$ under consistency. Define

$$\tau = \mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)], \tag{1}$$
$$\mathrm{ATT} = \mathbb{E}[Y(1) - Y(0) \mid A = 1], \tag{2}$$
$$\mathrm{ATC} = \mathbb{E}[Y(1) - Y(0) \mid A = 0]. \tag{3}$$

Writing $p = \mathbb{P}(A = 1)$, the law of total expectation gives

$$\mathrm{ATE} = p\,\mathrm{ATT} + (1 - p)\,\mathrm{ATC}\,. \tag{4}$$

These are population targets, not estimators. Their numerical values need not agree in an observational study.

For a binary outcome, classify units by their pair of potential outcomes:

Outcome response type	$Y(0)$	$Y(1)$	Effect
Always has outcome	1	1	0
Never has outcome	0	0	0
Helped / responder	0	1	1
Harmed	1	0	-1

Thus

$$\mathbb{E}[Y(1)] = \mathbb{P}(\text{always}) + \mathbb{P}(\text{helped}), \tag{5}$$
$$\mathbb{E}[Y(0)] = \mathbb{P}(\text{always}) + \mathbb{P}(\text{harmed}), \tag{6}$$
$$\tau = \mathbb{P}(\text{helped}) - \mathbb{P}(\text{harmed}), \tag{7}$$
$$\mathrm{ATT} = \mathbb{P}(\text{helped} \mid A = 1) - \mathbb{P}(\text{harmed} \mid A = 1). \tag{8}$$

Response types versus compliance types

The manuscript calls the first two rows "always-taker" and "never-taker." Here the table concerns outcomes, so "always has outcome" and "never has outcome" are more precise. IV compliance types instead concern potential *treatment uptake*, $A(0), A(1)$; see Note 5.

Question in the margin: what if the harmed proportion is known?

The identity for τ is valid irrespective of whether either response proportion is known. Knowing the harmed proportion alone does not identify the helped proportion. If randomization identifies τ and the harmed proportion is supplied separately, then

their sum identifies the helped proportion. Even an RCT generally identifies only the marginal distributions of $Y(0)$ and $Y(1)$, not their joint distribution.

2 SUTVA, consistency, and randomization

Manuscript: stat566-note01, pp. 2–4.

SUTVA (stable unit treatment value assumption) packages no interference and the absence of relevant hidden treatment versions. The course also spells out consistency separately.

1. **Treatment variation irrelevance.** If versions k and k' represent the same treatment level a, then $Y_i(a,k) = Y_i(a,k')$. Different doses should not be silently treated as equivalent versions if they change the outcome.
2. **Consistency.** If $A_i = a$, then $Y_i = Y_i(a)$. Equivalently,
$$Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0). \tag{9}$$
3. **No interference.** A unit's potential outcome depends on its own treatment, not the treatment assignments to other units:
$$Y_i(a_1, \dots, a_n) = Y_i(a_i). \tag{10}$$
Vaccination is a counterexample when other people's vaccination protects an untreated person.

TODO / questions, p. 3: consistency and "EQC?"

The original asks to reread the reference and contains the shorthand "EQC?" twice; that abbreviation cannot be reliably expanded. The relevant equality is

$$\mathbb{E}[Y(1) \mid A = 1] = \mathbb{E}[Y \mid A = 1], \tag{11}$$

which follows from consistency alone. In contrast, replacing $\mathbb{E}[Y(1) \mid A = 1]$ by $\mathbb{E}[Y(1)]$ needs exchangeability. The ambiguous abbreviation is preserved here rather than guessed.

In a randomized experiment, treatment assignment is independent of potential outcomes (and baseline covariates under simple randomization):

$$A \perp\!\!\!\perp (Y(0), Y(1), Z). \tag{12}$$

Consequently, for $a = 0, 1$ with positive assignment probability,

$$\mathbb{E}[Y(a)] = \mathbb{E}[Y(a) \mid A = a] = \mathbb{E}[Y \mid A = a], \tag{13}$$
$$\tau = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]. \tag{14}$$

Population ATE, ATT, and ATC then agree. A realized randomized sample need not have exact balance: baseline summary statistics diagnose chance imbalance but do not prove or disprove the assignment mechanism.

Consistency is not independence from intention

Consistency is the equality $Y = Y(A)$, not an independence statement about a person's intention to receive treatment. Random assignment of an offer identifies an intention-to-treat effect; noncompliance creates a separate question about actual treatment receipt.

3 Neyman estimation and Fisher randomization inference

Manuscript: stat566-note01, pp. 4–5; Lecture 1, randomization inference.

For a completely randomized assignment of n_1 treated and $n_0 = n - n_1$ controls, the difference in sample means estimates the finite-population ATE:

$$\hat{\tau} = \overline{Y}_1 - \overline{Y}_0 = \frac{1}{n_1} \sum_{i:A_i=1} Y_i - \frac{1}{n_0} \sum_{i:A_i=0} Y_i, \tag{15}$$

$$\tau_n = \frac{1}{n} \sum_{i=1}^{n} \{Y_i(1) - Y_i(0)\}, \qquad \mathbb{E}_A[\hat{\tau}] = \tau_n. \tag{16}$$

Here the potential outcomes are fixed; randomness is over the treatment assignments. Define finite-population variances using all n units:

$$S_a^2 = \frac{1}{n-1} \sum_{i=1}^{n} \{Y_i(a) - \overline{Y}(a)\}^2, \tag{17}$$

$$S_\tau^2 = \frac{1}{n-1} \sum_{i=1}^{n} \{Y_i(1) - Y_i(0) - \tau_n\}^2. \tag{18}$$

Then

$$\mathrm{Var}_A(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}. \tag{19}$$

The within-arm sample variances s_a^2 give the usual conservative estimator

$$\hat{V} = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}, \qquad \mathbb{E}_A[\hat{V}] = \mathrm{Var}_A(\hat{\tau}) + \frac{S_\tau^2}{n}. \tag{20}$$

Variance notation and the asymptotic pivot

The manuscript mixes sample variances computed within the observed arms with the finite-population variances in Neyman's exact formula. Moreover, $\hat{V} \geq \mathrm{Var}_A(\hat{\tau})$ is not guaranteed for every realized sample; conservativeness is in expectation. A central limit theorem concerns the centered statistic $(\hat{\tau} - \tau_n)/\sqrt{\mathrm{Var}_A(\hat{\tau})}$. Using $\hat{V}$ typically gives conservative large-sample confidence intervals under suitable regularity conditions, rather than an automatically exact standard normal pivot.

Fisher's sharp null is

$$H_0^{\text{sharp}}: \quad Y_i(1) = Y_i(0) \quad \text{for every } i. \tag{21}$$

It implies $\tau_n = 0$, but the converse fails: positive and negative individual effects can average to zero.

TODO, p. 5: reread Fisher's exact test / fill notes up to 10:08

Under the sharp null every missing potential outcome is imputed by the observed outcome. Choose a statistic such as $T = \overline{Y}_1 - \overline{Y}_0$, enumerate assignments allowed by the original design, recompute T for each, and calculate the fraction at least as extreme as the observed statistic:

$$p_{\text{exact}} = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathbf{1}\{|T(a, Y)| \geq |T(A^{\text{obs}}, Y)|\}. \tag{22}$$

This expression assumes equally likely assignments. Use design probabilities otherwise. Monte Carlo sampling of assignments approximates enumeration. The timestamp identifies a missing recording segment; its exact contents are unavailable, so this box supplies the lecture's core method rather than reconstructing an unheard segment.

4 Partial identification and Manski bounds

Manuscript: stat566-note01, pp. 6–7 (April 7); Lecture 2, Sections 1–2.

Identification asks what is determined by the population observed-data law; estimation approximates that quantity from a sample. Without exchangeability, the ATE need not be point identified. Assume both potential outcomes lie in a known interval $[L, U]$, and write $\mu_a = \mathbb{E}[Y \mid A = a]$, $p = \mathbb{P}(A = 1) \in (0, 1)$, and $q = 1 - p$. Consistency gives

$$\mathbb{E}[Y(1)] = p\mu_1 + q\mathbb{E}[Y(1) \mid A = 0], \tag{23}$$
$$\mathbb{E}[Y(0)] = q\mu_0 + p\mathbb{E}[Y(0) \mid A = 1]. \tag{24}$$

The missing counterfactual means can range over $[L, U]$, so

$$p\mu_1 + qL \leq \mathbb{E}[Y(1)] \leq p\mu_1 + qU, \tag{25}$$
$$q\mu_0 + pL \leq \mathbb{E}[Y(0)] \leq q\mu_0 + pU. \tag{26}$$

Subtract opposite endpoints to obtain sharp bounds:

$$\tau_L = p(\mu_1 - U) + q(L - \mu_0), \tag{27}$$
$$\tau_U = p(\mu_1 - L) + q(U - \mu_0), \tag{28}$$
$$\tau \in [\tau_L, \tau_U], \qquad \tau_U - \tau_L = U - L. \tag{29}$$

Sharp means every endpoint is attainable by some potential-outcome distribution consistent with the observed law and maintained assumptions. Bounded support alone gives $[L - U, U - L]$, of width $2(U - L)$; observing treatment and outcome cuts that width in half.

Questions, p. 7: can the lower endpoint be zero?

Yes, $L = 0$ is allowed; use whatever support is substantively justified. For binary outcomes $[L, U] = [0, 1]$, the trivial ATE range is $[-1, 1]$, but Manski's data-dependent interval has width 1. For $p = .4$, $\mu_1 = .7$, $\mu_0 = .5$, the interval is $[-.42, .58]$. The naive contrast .2 does not identify the ATE. A bounded *observed* outcome alone does not establish bounds on unobserved potential outcomes without the corresponding maintained support restriction.

5 Monotone treatment response and selection

Manuscript: stat566-note01, pp. 8–9; Lecture 2, Section 3.

MTR (monotone treatment response) assumes $Y_i(1) \geq Y_i(0)$ for every unit. If larger values are beneficial, it rules out harm. This is a substantive restriction, not a general property of treatments. It implies

$$\mathbb{E}[Y(1) \mid A = 0] \geq \mu_0, \tag{30}$$
$$\mathbb{E}[Y(0) \mid A = 1] \leq \mu_1, \tag{31}$$
$$\tau \in [0, \tau_U]. \tag{32}$$

TODO, p. 8: complete the MTR bound

The upper endpoint remains $p(\mu_1 - L) + q(U - \mu_0)$; it is generally not simply U. A zero effect for every unit achieves the lower endpoint while respecting each observed outcome. Setting untreated units' missing treated outcomes to U and treated units' missing control outcomes to L achieves the upper endpoint and respects MTR.

Positive MTS (monotone treatment selection) assumes

$$\mathbb{E}[Y(t) \mid A = 1] \geq \mathbb{E}[Y(t) \mid A = 0], \qquad t = 0, 1. \tag{33}$$

It describes positive sorting into treatment on potential outcomes, not nonnegative treatment

effects. Thus

$$\mathbb{E}[Y(1) \mid A = 0] \leq \mu_1, \qquad \mathbb{E}[Y(0) \mid A = 1] \geq \mu_0, \tag{34}$$

$$\tau \in [\tau_L, \mu_1 - \mu_0] \quad \text{under MTS alone,} \tag{35}$$

$$\tau \in [0, \mu_1 - \mu_0] \quad \text{under MTR and MTS.} \tag{36}$$

The lower bound zero needs MTR

The manuscript's final interval implicitly combines MTR and MTS. Positive MTS alone does not prevent negative causal effects. If $\mu_1 < \mu_0$, the combined assumptions are incompatible with the observed law. Some settings motivate negative rather than positive selection; the direction must be justified. These are identification bounds, not confidence intervals for estimated endpoints.

Lecture cross-reference

Ema Perković, STAT 566, Spring 2026: Lecture 1, *Introduction and Randomized Experiments* (SUTVA, Neyman estimation, Fisher inference); Lecture 2, *Manski Bounds and Partial Identification*, Sections 1–3. Local source filenames and full manuscript coverage are listed in `SOURCE-MAP.md`.